Data and People

Definition

SENSITIVE ATTRIBUTE : data about people with **legal** or **ethical** implications (privacy, non-discrimination).

Key formulas

Legally sensitive

- **GDPR** : personal data (identity, directly or indirectly); **special categories banned by default** (ethnicity, politics, religion, health).
- **Non-discrimination law** : direct and **indirect** discrimination (apparently neutral practice, disadvantage for a **protected group**).

Ethically sensitive : not in the law but can lead to discrimination (height).

Definition

PROXY ATTRIBUTE : correlates with a sensitive attribute $\Rightarrow$ indirect discrimination, **harder to detect** (SAT score $\sim$ race).

Key idea

Observed vs. **latent** variables : arrests for crime, clicks for interest.
Data $\neq$ **objectivity** : an imperfect representation, built from choices with a goal.

Example

Hair colour : not legally sensitive, **observed**, **subjective** (red vs. auburn).

Explained to a goat

Some facts about you are protected by law, some are not but can still be used against you, and innocent facts can secretly reveal protected ones.

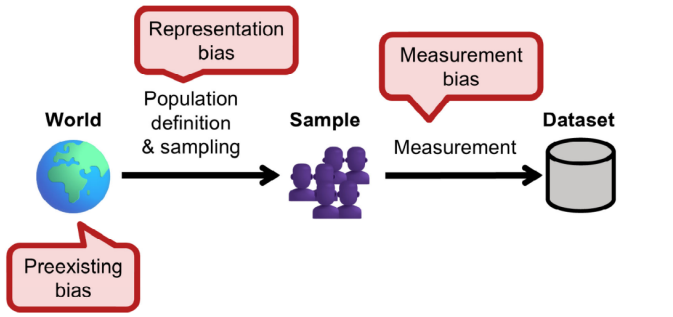
Sources of Unfairness

Definition

FAIRNESS : equality of **what** (resources, opportunities), **how** (same treatment vs. equity), for **whom**?
BIAS : systematic deviation from a standard; relative, not always bad; leads to **unfairness** (discrimination, errors, stereotyping). In **algorithms**, **data**, **people**.

Bias in data

(Adapted from Suresh & Guttag, 2021; Fazelpour & Danks, 2021; Mehrabi et al., 2021; Schwartz et al., 2022)



Key formulas

Bias in data (Suresh & Guttag)

- **Pre-existing** : the world is unfair and the data records it (systemic).
- **Representation** : some people missing or under-represented (sampling, population).
- **Measurement** : inaccurate values, worse for some groups.

Example

Pothole app $\rightarrow$ **representation**. Same calorie formula for walking and running $\rightarrow$ **measurement**.

Explained to a goat

Unfair world (pre-existing), uncounted goats (representation), a scale wrong for some goats (measurement).

People and Bias

Definition

COGNITIVE BIASES : systematic deviations from rationality from System 1 shortcuts; inherent, unconscious, unrelated to intelligence. Affect users and designers.

Key formulas

- **Implicit stereotypes** : "default user", harmful stereotypes.
- **Confirmation bias** : keep evidence that fits your belief; design fixation.
- **Group effects** : homogeneous groups, **groupthink**.
- **Sunk cost fallacy** : continue because of past investment; ethical debt.

Key idea

Improve : **know** your biases (IAT), **resist** (slow down, structure), **involve** diverse people.

Example

Five similar engineers keep only the feedback supporting their idea : **confirmation bias**.

Explained to a goat

The brain imagines a "normal" user, believes itself, agrees with similar friends and hates dropping a paid-for plan.

Strategy : Inclusive Design

Definition

INCLUSIVE DESIGN : considers the **full range of human diversity** (ability, language, culture, gender, age…). **Disability = mismatch** between system and needs.

Key formulas

01 **Capabilities** needed for the app **and the task** (table : perceive, think, act).

02 **Non-average users** for each capability.

03 **Expand** : other users, non-users, minorities (not from the table).

04 **Solutions** for each NAU.

Stage 01: needed capabilities

Perceive	Vision	Detect objects, discriminate surfaces, etc.
	Hearing	Discriminate tones/speech, localize sounds, etc.
	Touch & Haptic	Discriminate pressures, detect temperature, etc.
	Smell	Detect odors, discriminate odors, etc.
	Taste	Detect 5 basic tastes, detect temperature, etc.
Think	Thinking	Process info., attention, memories, select appropriate responses & actions, etc.
	Communicate	Understand others, express oneself, etc.
Act	Locomotion	Move around, bend down, climb steps, etc.
	Reach & Stretch	Put arm(s) out in front of body/above head/behind back, etc.
	Dexterity	Fine finger manipulation, pick up, carry/grasp/squeeze objects, etc.

Example

Colour blindness → "EVEN"/"ODD" text; deafness → visual feedback; ADHD → no animated goalkeepers; dyslexia → special font; plaster hand → larger hitboxes; low digital experience → tutorial.

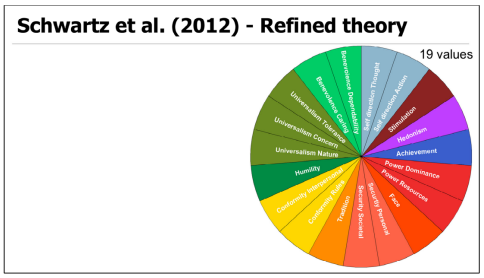
Explained to a goat

What does the app ask of eyes, ears, mind and hands? Who cannot? Change the design so they can.

Strategy : Value Analysis

Definition

VALUES (Schwartz) : transsituational goals guiding a person's life. 19 values in 4 groups : **self-enhancement** (power, achievement, face), **openness to change** (hedonism, stimulation, self-direction action / thought), **self-transcendence** (universalism, benevolence, humility), **conservation** (tradition, conformity, security).



Key formulas

Step 01 : table stakeholder / key value / manifested / **benefit** / **harm**.
Step 02 : **tension map** : values, stakeholders + / −, lines between a benefit and a harm (between or within stakeholders, same or different values).

Example

Smart diaper : parents **+** self-direction action (decide feeding time) vs. parents − self-direction thought (lose parenting skills).

Key idea

Products are **not neutral** : they embed values. Seeing tensions is the first step.

Explained to a goat

Your app helps some things people care about and hurts others; draw who wins and who loses.

Weekly Summary

Summary

- Sensitive (legal / ethical) and proxy attributes; data ≠ objectivity.
- Fairness : what, how, whom. Bias → unfairness.
- Data bias : pre-existing, representation, measurement.
- Human biases : stereotypes, confirmation, group effects, sunk cost.
- Inclusive design : 4 stages. Value analysis : table + tension map.